

Bipartite mutual information of flattened domain-wall ground states

Internal campaign summary¹

¹ *Class-A fermionic adaptive-circuit project*

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We compute deterministic reference curves for the half-filled ground state of the overcomplete-Wannier flattened parent associated with a class-A adaptive Chern circuit. The spatial geometry is a periodic $N_x \times N_y$ torus with $N_x = 20$, two domain walls, and two opposite full- x strips of width $w = \lfloor N_y/4 \rfloor$. For $N_y = 20, 24, 28$ at $n_{\text{shell}} = 1$, and for $N_y = 40, 50, 60$ at $n_{\text{shell}} = 1, 2, \infty$, we evaluate the strip mutual information and the finite-size entropy slope for hard support-truncated and soft untruncated wall constructions across a 21-point sweep from $\alpha_1 = 3$ to 1. In the topological regime the mutual information approaches the free-Dirac fixed-cross-ratio value $(\log 2)/3 = 0.2310491$ for exact quarter strips, while the entropy slope gives $c_{\text{fit}} \simeq 1$. The mutual information does not diverge under this finite-size scaling because strip width and separation grow in the same proportion, leaving the conformal cross-ratio fixed. The full-frame enhancement just below $\alpha_1 = 2$ lies in a bulk-critical finite-size window, but it does not coincide with the smallest saved half-filling gap; it is not evidence for a stable wall theory with central charge greater than one. These results are exact-Hamiltonian benchmarks, not results of the planned Born-trajectory dynamical sweep.

CONTENTS

I. Scope and principal conclusions	1
II. Flattened OW parent and domain walls	1
A. Uniform parent and OW frame	1
B. Hard and soft interfaces	2
C. Campaign matrix and deterministic solver	2
D. Critical-grid prescription	2
III. Gaussian entropy and estimators	2
IV. Two-interval mutual information in the Dirac CFT	3
A. Line result	3
B. Circle geometry and the quarter-strip value	4
C. When does critical mutual information diverge?	4
V. Numerical results	4
A. Finite support: small circumferences	4
B. Large circumferences and shell dependence	5
VI. Validation and limitations	6
VII. Conclusion	9
References	10

than a simulation of that circuit: we form the quadratic OW flattened parent, fill its lowest half of single-particle levels, and measure entanglement in the resulting Slater determinant. There are consequently no Born outcomes, stochastic trajectories, time evolution, or sampling error bars in this note.

Three conclusions survive changes in circumference and wall construction. First, in the topological regime the opposite-strip mutual information tends to the free-Dirac value at the chosen cross-ratio, while the single-strip entropy has a logarithmic coefficient consistent with one nonchiral Dirac channel. Second, shortening the OW support displaces and broadens the finite-size crossover relative to the untruncated parent transition at $\alpha_1 = 2$. Third, the sharp enhancement of the full-frame estimators just below $\alpha_1 = 2$ occurs where the bulk correlation length contaminates a wall-only fit. The saved half-filling diagnostics do not support attributing that enhancement to a zero-mode degeneracy, and it must be separated from the stable $c_{\text{fit}} \simeq 1$ plateau at smaller α_1 .

All entropies and mutual informations below use natural logarithms and are therefore measured in nats. The production arrays, CSV tables, and their checksum-bound completion records remain the authoritative numerical products; the figures in this note are a read-only rendering of those arrays.

II. FLATTENED OW PARENT AND DOMAIN WALLS

A. Uniform parent and OW frame

The auxiliary Bloch Hamiltonian is the two-band matrix

$$h_\alpha(\mathbf{k}) = \hat{\mathbf{n}}_\alpha(\mathbf{k}) \cdot \boldsymbol{\sigma}, \quad (1)$$

$$\mathbf{n}_\alpha(\mathbf{k}) = (\sin k_x, \sin k_y, \alpha - \cos k_x - \cos k_y), \quad (2)$$

I. SCOPE AND PRINCIPAL CONCLUSIONS

The class-A adaptive protocol of Bhuiyan, Pan, and Jian (BPJ) constructs localized, nonorthogonal overcomplete Wannier (OW) modes from a two-band Chern parent and uses their occupation measurements as a local controller [1]. A spatially patterned controller creates two parallel domain walls on a periodic system. The purpose of the calculation reported here is deliberately narrower

Away from a band closing, the spectral projectors are $P_{\pm}(\mathbf{k}) = [\mathbf{1} \pm h_{\alpha}(\mathbf{k})]/2$. Projecting two trial orbitals τ_{ν} , $\nu = A, B$, into the two bands and Fourier transforming produces the OW states $|W_{\mathbf{r},\nu,\pm}\rangle$. They are normalized after the real-space support and wall masks are applied, but they are neither orthogonal nor a linearly independent Wannier basis. Their frame matrices are

$$F_{\pm} = \sum_{\mathbf{r},\nu=A,B} |\widetilde{W}_{\mathbf{r},\nu,\pm}\rangle \langle \widetilde{W}_{\mathbf{r},\nu,\pm}|. \quad (3)$$

The single-particle matrix and number-conserving many-body Hamiltonian used in this benchmark are

$$h_{\text{flat}} = F_+ - F_-, \quad (4)$$

$$\hat{H}_{\text{flat}} = \sum_{i,j} \hat{c}_i^{\dagger} (h_{\text{flat}})_{ij} \hat{c}_j. \quad (5)$$

Here “flattened” refers to the signs assigned to the upper- and lower-band OW frame contributions. Because the OW frame is overcomplete, h_{flat} need not itself have only the eigenvalues ± 1 .

For each configuration we diagonalize h_{flat} and occupy its lowest $N/2 = N_x N_y$ single-particle eigenvectors, where $N = 2N_x N_y$ includes both orbitals. If V_{occ} denotes the matrix of occupied vectors, the ground-state correlation projector is

$$C = V_{\text{occ}} V_{\text{occ}}^{\dagger}, \quad C^2 = C = C^{\dagger}. \quad (6)$$

B. Hard and soft interfaces

The fixed transverse size is $N_x = 20$. The topological controller occupies $x = 5, \dots, 15$ and uses α_1 ; the exterior uses $\alpha_2 = 30$. Thus the two interfaces are at $x = 5$ and $x = 16$, as shown in Fig. 1. For the hard wall, an OW mode is multiplied by a domain mask that forbids its support from crossing either interface before the mode is normalized. For the soft wall, this additional mask is absent: the parent parameter is still selected by the mode center, but the resulting mode may extend across an interface. The shell mask restricts periodic coordinate differences to at most n_{shell} cells in each direction; $n_{\text{shell}} = \infty$ retains the full OW mode.

C. Campaign matrix and deterministic solver

Table I summarizes the two completed CPU calculations. Both use complex128 arithmetic, exact half filling, the X trial-orbital pair, and the same descending α_1 grid,

$$\{3, 2.75, 2.5, 2.3, 2.2, 2.15, 2.1, 2.075, 2.05, 2.025, 2, 1.975, 1.95, 1.925, 1.9, 1.85, 1.8, 1.7, 1.5, 1.25, 1\}. \quad (7)$$

The first campaign performs dense Torch diagonalization. The second exploits exact translation symmetry along y :

Fourier transformation reduces h_{flat} to N_y independent blocks of dimension $2N_x = 40$, which are diagonalized with NumPy/SciPy. Filling the lowest half of the combined block spectrum is equivalent to diagonalizing the full matrix.

D. Critical-grid prescription

The allowed periodic momentum grid contains $\mathbf{k} = \mathbf{0}$. At the uniform parent transition $\alpha = 2$, Eq. (2) gives $\mathbf{n}_2(\mathbf{0}) = \mathbf{0}$, so $\hat{\mathbf{n}}_2(\mathbf{0})$ and the upper/lower band decomposition are not uniquely defined. The completed implementation uses the symmetric numerical prescription

$$h_2(\mathbf{0}) = 0, \quad P_+(\mathbf{0}) = P_-(\mathbf{0}) = \frac{1}{2}. \quad (8)$$

The two matrices in Eq. (8) resolve the identity but are not idempotent spectral projectors. Accordingly, every plotted $\alpha_1 = 2$ point is a finite-grid midpoint convention, not either one-sided limit of the gapped band projector. Open markers identify those points in the result figures. Plateau and crossover claims are based on the neighboring gapped values rather than on the precise midpoint value.

III. GAUSSIAN ENTROPY AND ESTIMATORS

For a number-conserving Gaussian state, the reduced density matrix of a region R is fixed by the restricted correlation matrix C_R . If its eigenvalues are $\lambda_q \in [0, 1]$, the von Neumann entropy is

$$S_R = -\text{Tr} [C_R \log C_R + (\mathbf{1}_R - C_R) \log(\mathbf{1}_R - C_R)], \quad (9)$$

which is evaluated by diagonalizing C_R [2, 3]. Numerically, eigenvalues are clipped only inside the logarithm at 10^{-12} ; the unclipped extrema are retained as diagnostics.

For every allowed translation y_0 , define

$$a(y_0) = [0, N_x) \times [y_0, y_0 + w), \quad (10)$$

$$b(y_0) = [0, N_x) \times [y_0 + N_y/2, y_0 + N_y/2 + w), \quad (11)$$

with periodic y wrapping and both orbitals included. The nonlinear observable is formed before translation averaging,

$$I_{a,b}(y_0) = S_{a(y_0)} + S_{b(y_0)} - S_{a(y_0) \cup b(y_0)}. \quad (12)$$

The small-size campaign explicitly averages Eq. (12) over $N_y/2$ placements; the omitted half exchanges a and b . Its largest spread over y_0 is 2.74×10^{-12} . The large-size solver uses exact y -translation symmetry to evaluate one placement and verifies a shift by one lattice row, for which all recorded entropy and mutual-information differences vanish at saved precision.

The single-strip entropy profile uses widths $A_y = 1, \dots, N_y/2$. We determine an operational central-charge

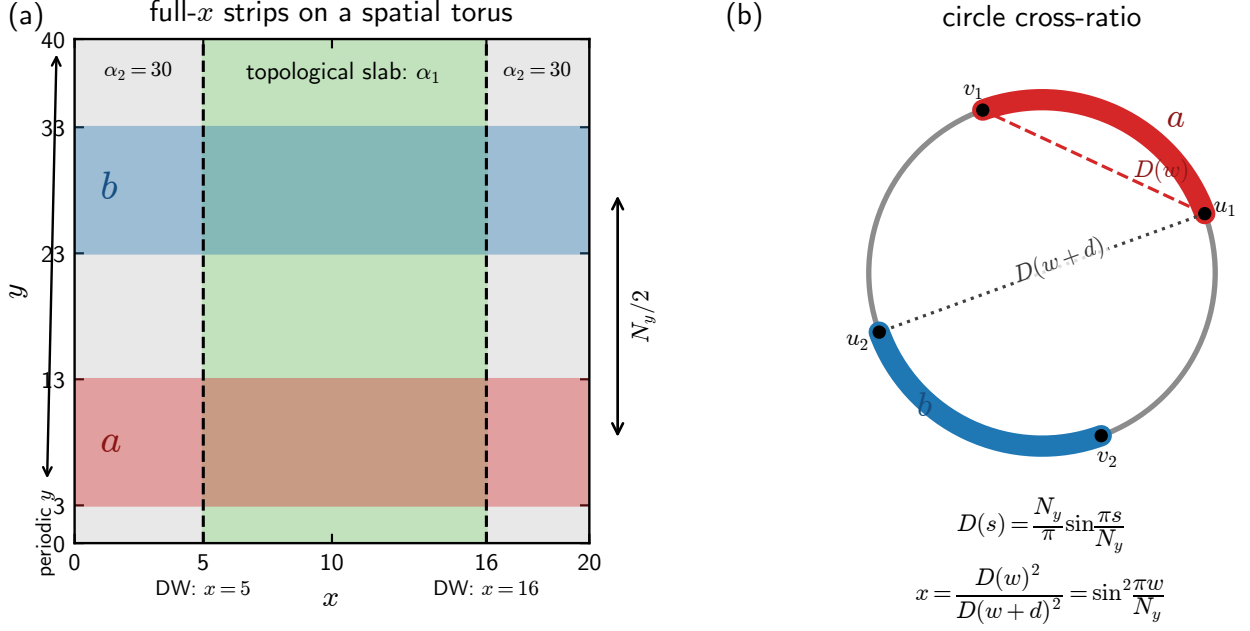


FIG. 1. Mutual-information geometry and its one-dimensional CFT reduction. (a) The full- x regions a and b include both orbitals and cross both domain walls. Their starting points differ by $N_y/2$ on the periodic y circle. The drawing uses the representative $N_x = 20$, $N_y = 40$ case; a and b each have width $w = N_y/4 = 10$. The topological slab comprises sites $x = 5, \dots, 15$, so its interfaces lie at $x = 5$ and $x = 16$. (b) The wall degrees of freedom see two equal intervals on a circle. Conformal mapping replaces a line distance s by its chord distance $D(s) = (N_y/\pi) \sin(\pi s/N_y)$. For opposite intervals, $w + d = N_y/2$, and the cross-ratio reduces to $x = D(w)^2/D(w+d)^2 = \sin^2(\pi w/N_y)$.

TABLE I. Deterministic flattened-ground-state campaigns. Each listed point is one Hamiltonian ground state rather than an ensemble of trajectories.

campaign	N_x	N_y	w	n_{shell}	walls	cases
small circumference	20	20, 24, 28	5, 6, 7	1	hard, soft	126
large circumference	20	40, 50, 60	10, 12, 15	1, 2, ∞	hard, soft	378

estimator from the unweighted fit

$$S(A_y) = m \log \left[\frac{N_y}{\pi} \sin \left(\frac{\pi A_y}{N_y} \right) \right] + b, \quad (13)$$

$$c_{\text{fit}} = 3m, \quad A_y = 2, \dots, N_y/2. \quad (14)$$

The chord-length dependence is the standard periodic CFT result [4]. The two spatially separated walls carry oppositely oriented chiral complex modes. Together they supply the left- and right-moving sectors of one nonchiral Dirac theory, so their combined coefficient is $m = c/3$ with $c = 1$. The saved fit standard error measures residual curvature about Eq. (14); it is not a statistical uncertainty.

IV. TWO-INTERVAL MUTUAL INFORMATION IN THE DIRAC CFT

A. Line result

The cancellation of local boundary-law terms explains why mutual information is ultraviolet finite for separated regions, but it does not by itself fix the dependence on interval geometry. That dependence follows here from a special exact property of the massless free Dirac field. Put two intervals on the line in the order

$$A = [u_1, v_1], \quad B = [u_2, v_2], \quad u_1 < v_1 < u_2 < v_2, \quad (15)$$

and define

$$\ell_a = v_1 - u_1, \quad d = u_2 - v_1, \quad \ell_b = v_2 - u_2. \quad (16)$$

Replica diagonalization and bosonization turn the free-fermion endpoint twist correlator into a product of pairwise endpoint separations. The resulting von Neumann

entropy is [5]

$$S_{A \cup B} = \frac{c}{3} \log \left[\frac{\ell_a \ell_b d(\ell_a + d + \ell_b)}{\epsilon^2 (\ell_a + d)(\ell_b + d)} \right] + 2s_0, \quad (17)$$

whereas the two single-interval entropies give

$$S_A + S_B = \frac{c}{3} \log \left(\frac{\ell_a \ell_b}{\epsilon^2} \right) + 2s_0. \quad (18)$$

Both the cutoff ϵ and endpoint constant s_0 cancel, leaving

$$I_{a,b} = \frac{c}{3} \log \left[\frac{(\ell_a + d)(\ell_b + d)}{d(\ell_a + \ell_b + d)} \right]. \quad (19)$$

Introducing the cross-ratio

$$x = \frac{\ell_a \ell_b}{(\ell_a + d)(\ell_b + d)}, \quad 1 - x = \frac{d(\ell_a + \ell_b + d)}{(\ell_a + d)(\ell_b + d)}, \quad (20)$$

puts Eq. (19) in the compact form

$$I_{a,b} = -\frac{c}{3} \log(1 - x). \quad (21)$$

Equation (21) is not fixed by conformal symmetry and central charge alone. For a generic 1 + 1-dimensional CFT, the replica answer is a four-point function of twist fields and contains a theory-dependent scaling function $\mathcal{F}_n(x)$ [6]. Its $n \rightarrow 1$ derivative contributes an additional function of x to the von Neumann mutual information. The simple logarithm is exact for the massless free Dirac theory (and additive copies thereof); substituting a continuously fitted “effective central charge” in a more general crossover is only a scaling ansatz.

B. Circle geometry and the quarter-strip value

In the conformal-vacuum sector obtained by mapping the plane vacuum to a circle of circumference N_y , every endpoint separation s is replaced by

$$D(s) = \frac{N_y}{\pi} \sin \left(\frac{\pi s}{N_y} \right). \quad (22)$$

Common factors of N_y/π cancel in the cross-ratio. For two intervals with lengths ℓ_a, ℓ_b and intervening gap d ,

$$x = \frac{\sin(\pi \ell_a / N_y) \sin(\pi \ell_b / N_y)}{\sin[\pi(\ell_a + d) / N_y] \sin[\pi(\ell_b + d) / N_y]}. \quad (23)$$

Our intervals are equal, $\ell_a = \ell_b = w$, and their starting points are opposite. Hence $d = N_y/2 - w$, so both denominator factors in Eq. (23) equal one and

$$x = \sin^2 \left(\frac{\pi w}{N_y} \right), \quad (24)$$

$$I_{a,b}^{\text{CFT}} = -\frac{c}{3} \log \left[\cos^2 \left(\frac{\pi w}{N_y} \right) \right]. \quad (25)$$

For an exact quarter strip, $w = N_y/4$, this becomes

$$I_{a,b}^{\text{CFT}} = \frac{c}{3} \log 2. \quad (26)$$

The $c = 1$ prediction is 0.2310491. The only non-quarter geometry in the campaign is $N_y = 50$, where $w = \lfloor N_y/4 \rfloor = 12$; Eq. (25) then gives 0.2107497.

This circle formula is an exact statement for the corresponding free-Dirac conformal vacuum, not a sector-independent identity for every finite periodic fermion problem. Periodic (Ramond) boundary conditions can introduce zero modes and ground-state degeneracy, for which zero-mass and zero-temperature limits need not commute [7]. The lattice calculation uses periodic y boundary conditions and a definite half-filled projector; we therefore use Eqs. (25) and (26) as continuum benchmarks, while diagnosing finite-size zero modes separately.

C. When does critical mutual information diverge?

A critical phase has correlations on all length scales, but the mutual information of two separated regions need not diverge. In the present finite-size sequence,

$$\frac{w}{N_y} = \frac{d}{N_y} = \frac{1}{4} \quad (27)$$

for the exact-quarter cases. Increasing N_y rescales every endpoint distance by the same factor and leaves $x = 1/2$ invariant. Scale invariance therefore predicts the nonzero constant in Eq. (26).

A logarithmic divergence is obtained in a different geometric limit. On the line, take two equal intervals of length w separated by a fixed gap d . Equation (19) becomes

$$I_{a,b} = \frac{c}{3} \log \left[\frac{(w + d)^2}{d(2w + d)} \right] \simeq \frac{c}{3} \log \left(\frac{w}{2d} \right) \quad (w \gg d). \quad (28)$$

It also diverges when $d \rightarrow 0$, where the regions become adjacent and the short-distance cutoff reappears. Thus the plateau in this campaign is the expected critical signature for a fixed-cross-ratio geometry, not evidence against criticality.

V. NUMERICAL RESULTS

A. Finite support: small circumferences

Figure 2 shows the $n_{\text{shell}} = 1$ calculation for $N_y = 20, 24, 28$. At $\alpha_1 = 1$, hard-wall mutual informations range from 0.230112 to 0.230573, while soft-wall values range from 0.230908 to 0.230975. They lie within roughly 10^{-3} of $(\log 2)/3$. The corresponding central-charge fits range from 1.0033 to 1.0054. Both observables remain near their Dirac values through $\alpha_1 \simeq 1.5$; hard and soft walls agree closely throughout.

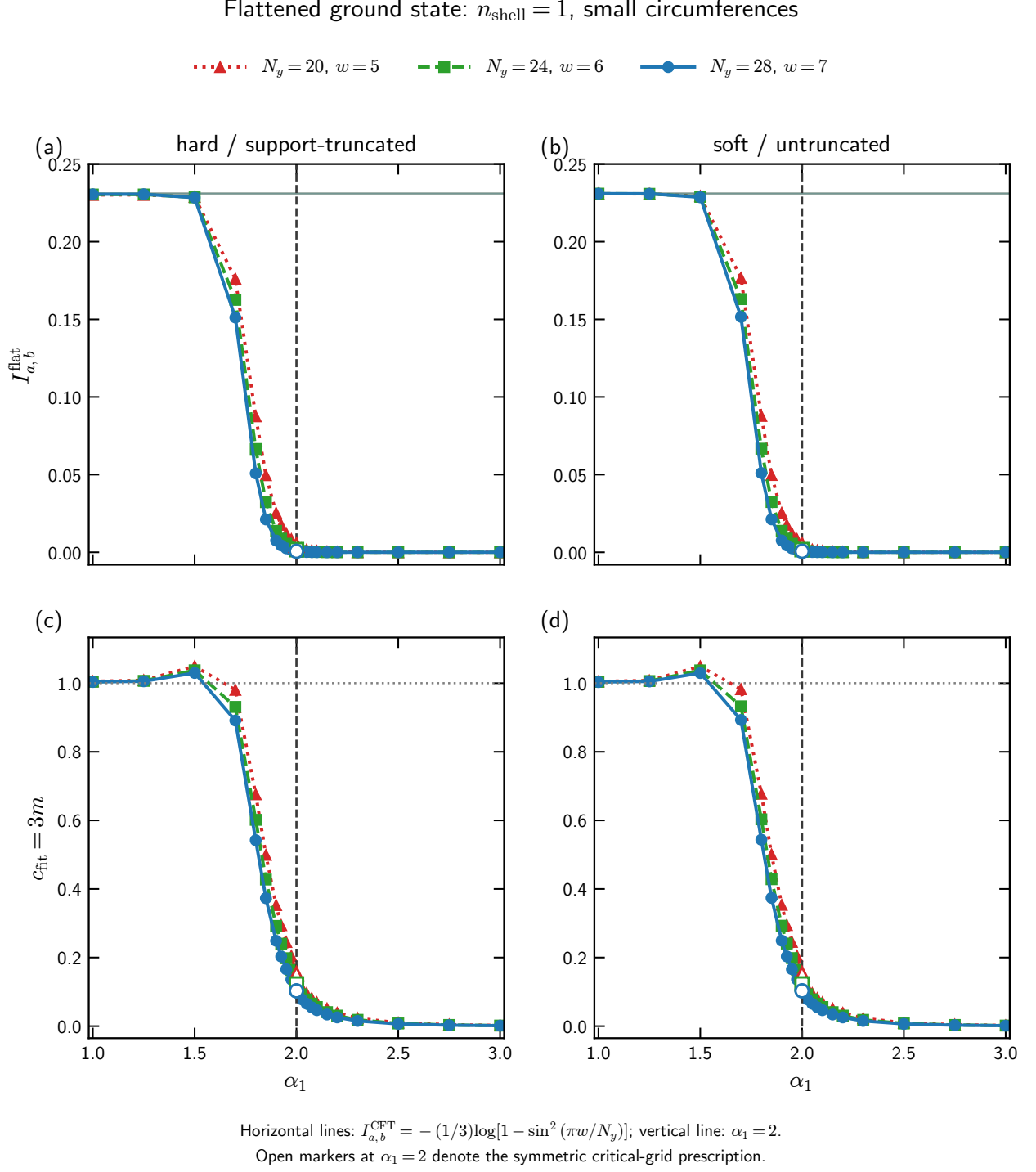


FIG. 2. Small-circumference flattened-ground-state sweep at $N_x = 20$, $N_y = 20, 24, 28$, $w = N_y/4 = 5, 6, 7$, $n_{\text{shell}} = 1$, and $\alpha_2 = 30$. Panels (a) and (b) show $I_{a,b}^{\text{flat}}$ for hard/support-truncated and soft/untruncated walls; panels (c) and (d) show $c_{\text{fit}} = 3m$ from $A_y = 2, \dots, N_y/2$. Colored horizontal lines in the upper row are the $c = 1$ free-Dirac references, which coincide at $(\log 2)/3$ for these exact quarter strips. The gray horizontal line is $c_{\text{fit}} = 1$, and the black dashed line marks $\alpha_1 = 2$; open markers there denote the symmetric prescription in Eq. (8). Every curve is deterministic and has no sampling uncertainty; the saved regression errors are fit-residual diagnostics, not ensemble error bars. Cases were evaluated on the descending grid $\alpha_1 : 3 \rightarrow 1$, while the display uses the conventional increasing axis.

B. Large circumferences and shell dependence

$n_{\text{shell}} = 1$, the exact-quarter values for $N_y = 40, 60$ lie

Figures 3–5 separate the larger systems by OW range so that every panel remains readable. At $\alpha_1 = 1$ and

between 0.230817 and 0.231033; the $N_y = 50$ values lie between 0.210601 and 0.210726, in accord with its distinct $w = 12$ reference. The fitted central charges are 1.0014–1.0023. Increasing to $n_{\text{shell}} = 2$ retains the same mutual-information plateau and gives modest positive finite-size curvature in c_{fit} , while moving the crossover closer to $\alpha_1 = 2$.

The $n_{\text{shell}} = \infty$ data behave differently only in the narrow bulk-critical region. Just below $\alpha_1 = 2$, the largest observed values are $I_{a,b} = 0.386194$ and $c_{\text{fit}} = 1.48664$ for the hard wall at $N_y = 40$ and $\alpha_1 = 1.975$. The half-filling gap at that same point is 0.176106, and the number of negative levels equals the imposed rank, 800. For the soft wall at the same size and parameter, the gap is 0.446436. Thus the saved half-filling spectrum does not show a zero-mode degeneracy at the overshoot.

The minimum gap across the campaign, 6.56×10^{-16} , and the only negative-level/rank mismatch, 799 versus 800, instead occur for the hard wall at $N_y = 40$ and $\alpha_1 = 1$, where the mutual information is on its ordinary topological plateau. This isolated wall-mode degeneracy is not evidence for the critical-window enhancement. The conservative explanation of that enhancement is that Eq. (14) assumes the only logarithmic sector is the one-dimensional wall theory, whereas the two-dimensional bulk correlation length grows rapidly near $\alpha_1 = 2$ and contaminates the finite-window fit. The overshoot is consequently a crossover diagnostic rather than a stable $c > 1$ phase. Away from this window the large- N_y curves again approach the $c = 1$ plateau for $\alpha_1 < 2$ and zero mutual information in the trivial regime. Hard and soft constructions agree most closely for finite shell and show their largest microscopic difference in the full-frame critical window.

VI. VALIDATION AND LIMITATIONS

The small aggregate contains arrays of shape (2, 3, 21) indexed by wall, circumference, and α_1 . The large aggregate has shape (2, 3, 3, 21), adding the shell index. Their completion records bind the NPZ byte counts and SHA-256 checksums. The manuscript renderer verifies those records before loading any array and then checks the wall labels, sizes, widths, shell labels, and the exact descending grid in Eq. (7).

The final occupied-state correlation projectors satisfy stringent numerical tests. For the small campaign the maximum projector idempotency residual is 2.89×10^{-15} and the maximum y -translation residual is 3.00×10^{-15} . For the large campaign the corresponding maxima are 5.44×10^{-15} and 1.04×10^{-14} for OW translation covariance. Restricted occupation eigenvalues remain within 1.5×10^{-14} of $[0, 1]$. These checks concern the final half-filled correlation projector; they do not turn the midpoint matrices $P_{\pm}(\mathbf{0}) = \mathbf{1}/2$ at $\alpha_1 = 2$ into spectral projectors. The smallest saved mutual information is -2.31×10^{-12} , a cancellation-scale roundoff value that is retained rather

than clipped; no value is negative beyond the 10^{-8} materiality tolerance.

Several limitations are essential. First, N_x is fixed, so increasing N_y tests the circumference dependence of two fixed transverse domain walls rather than isotropic two-dimensional finite-size scaling. Second, c_{fit} is a finite-window estimator and is not controlled when bulk modes become critical. Third, the exact Dirac expression for disconnected intervals does not transfer unchanged to an interacting or otherwise generic CFT. Fourth, the finite periodic lattice state need not realize the conformal-vacuum spin sector without corrections. Finally, this ground-state reference does not establish that the Born-weighted endpoint ensemble of the adaptive dynamics reaches the same curve. That comparison requires the independent trajectory sweep, including the prescribed order of nonlinear averaging over placements and samples.

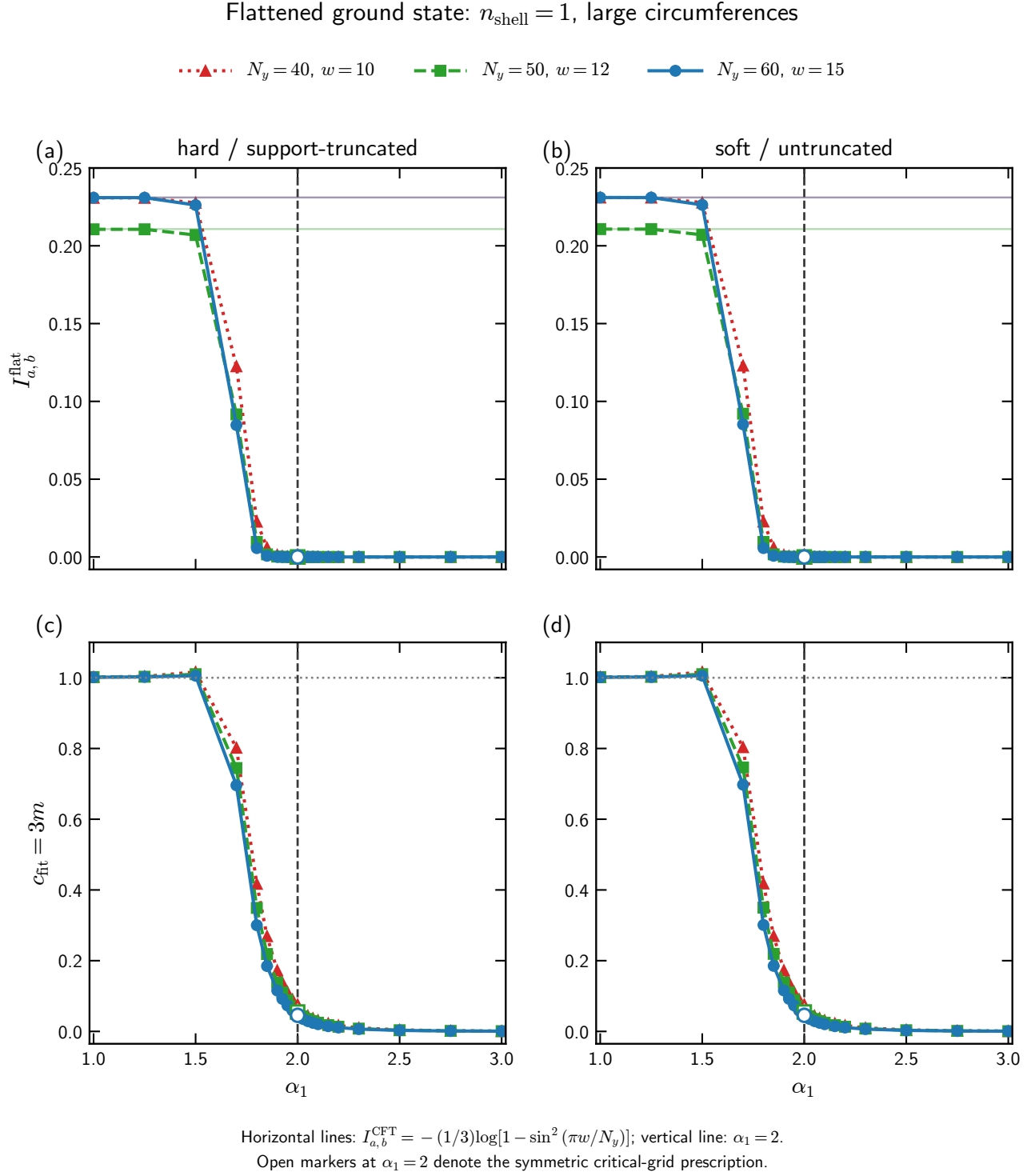


FIG. 3. Large-circumference sweep for $n_{\text{shell}} = 1$ at $N_x = 20$, $N_y = 40, 50, 60$, $w = 10, 12, 15$, and $\alpha_2 = 30$. Columns distinguish hard and soft walls; rows show mutual information and the entropy-slope estimator. The colored reference for $N_y = 50$ is 0.2107497 because $w = 12 < 50/4$; the $N_y = 40, 60$ references are $(\log 2)/3 = 0.2310491$. Open markers at $\alpha_1 = 2$ denote the symmetric critical-grid prescription in Eq. (8); line, marker, fit, determinism, and sweep conventions are otherwise the same as Fig. 2.

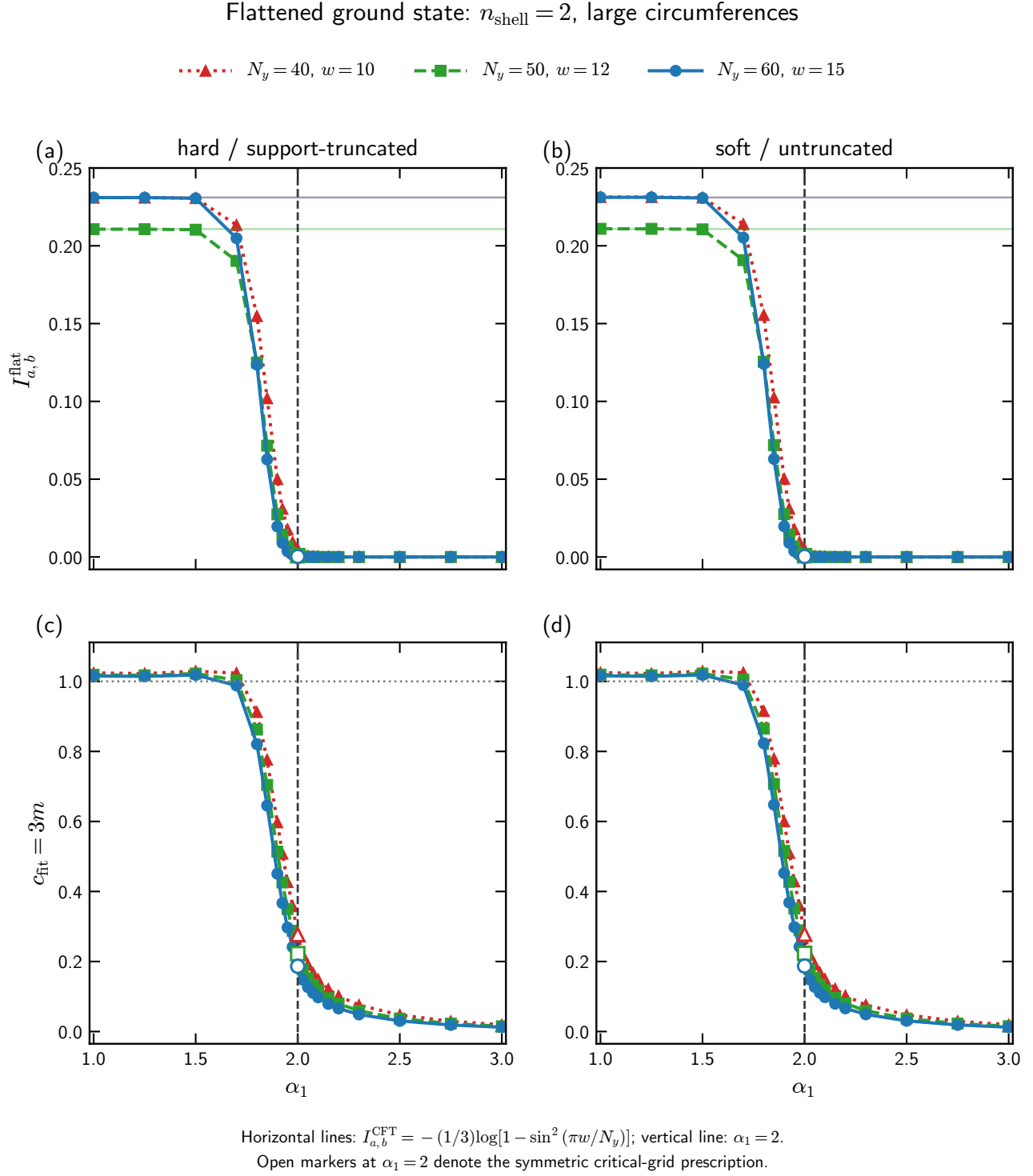


FIG. 4. Large-circumference sweep for $n_{\text{shell}} = 2$, with the same geometry and panel organization as Fig. 3. Extending the OW support preserves the topological mutual-information plateau but shifts the rounded crossover toward the untruncated parent value $\alpha_1 = 2$. The exact critical-grid points use Eq. (8). No sampling uncertainties are present; c_{fit} errors stored in the aggregate are residual errors of the chord-length regression. Cases were evaluated from $\alpha_1 = 3$ to 1 and are shown on an increasing horizontal axis.

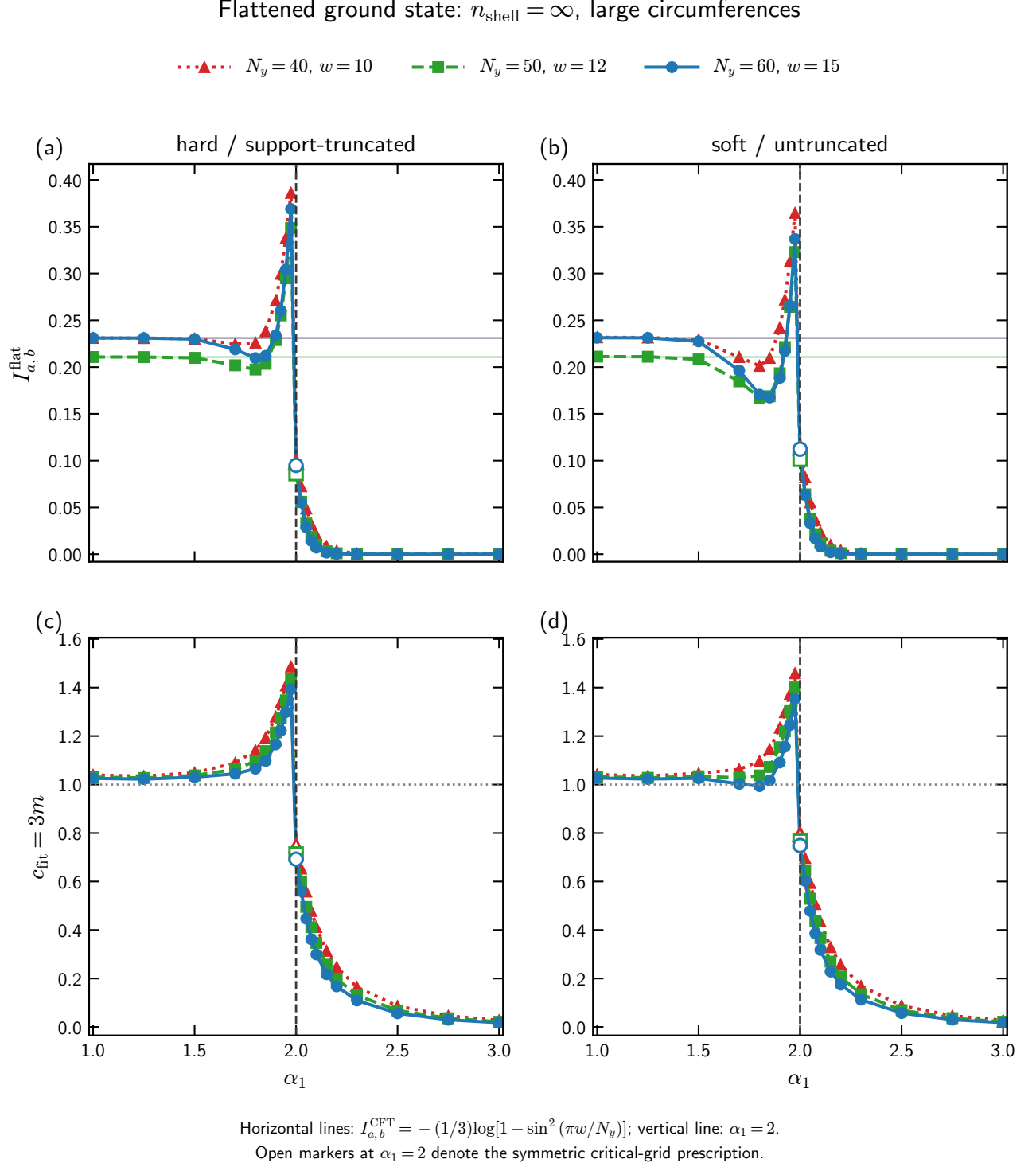


FIG. 5. Full-frame ($n_{\text{shell}} = \infty$) large-circumference sweep. The topological plateau at smaller α_1 remains close to the free-Dirac references. The enhancement immediately below $\alpha_1 = 2$ occurs where bulk criticality contaminates the wall-only chord fit. The saved half-filling gap at the enhancement is finite; the campaign’s isolated near-zero gap occurs instead at $\alpha_1 = 1$. The enhancement is therefore not interpreted as a stable $c > 1$ domain-wall theory or as a demonstrated zero-mode effect. Hard/soft columns, reference lines, deterministic status, regression window, and descending evaluation order are as in Figs. 2 and 3.

VII. CONCLUSION

The flattened OW parent supplies a clean calibration of the proposed opposite-strip observable. In the topological

regime, hard and soft walls and all studied circumferences recover a nonzero fixed-cross-ratio mutual- information

plateau together with a single-strip entropy coefficient near $c = 1$. The value $(\log 2)/3$ follows from the exact free-Dirac multi-interval identity and the quarter-circle geometry, not from area-law cancellation alone. Its size independence is precisely what scale invariance predicts when interval width and separation grow together. The

same derivation also identifies the geometry needed to obtain divergent mutual information: keep a physical gap fixed while growing the intervals, or take the adjacent-interval limit. These distinctions make the present curves useful quantitative references for, but not substitutes for, the future dynamical ensemble.

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